1 (a) Write 1.2×10^5 as an ordinary number. (1)

(b) Write 0.003 in standard form. (1)

(2 marks)

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2 (a) Write 42 900 000 in standard form. (1)

(b) Write 3.61×10^{-3} as an ordinary number. (1)

(2 marks)

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3 (a) Write 9.516×10^6 as an ordinary number. (1)

(b) Write 0.0724 in standard form. (1)

(c) Calculate $(8.694 \times 10^2) \div (6.21 \times 10^{-3})$
Give your answer in standard form. (2)

(4 marks)

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4 (a) Write 5.12×10^{-5} as an ordinary number. (1)

(b) Write 5 600 000 in standard form. (1)

(2 marks)

---

5 (a) Write 0.0065 in standard form. (1)

(b) Write 3×10^4 as an ordinary number. (1)

(2 marks)

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6 (a) Write 3.08×10^{-5} as an ordinary number. (1)

(b) Write 5 million in standard form. (1)

(c) Calculate $(6.3 \times 10^5) \times (2.5 \times 10^{-2})$
Give your answer in standard form. (2)

(4 marks)

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7 Work out $(8.69 \times 10^{-5}) \div (5.5 \times 10^{-7})$
Give your answer in standard form.

(2 marks)

---

8 (a) Write 0.00931 in standard form. (1)

(b) Write 7.429×10^3 as an ordinary number. (1)

(2 marks)

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9 (a) Write 5.2×10^{-1} as an ordinary number. (1)

(b) Work out the value of $(3.2 \times 10^3) \times (6.5 \times 10^4)$
Give your answer in standard form. (2)

(3 marks)

---

10 Write 0.21×10^6 in standard form.

(1 mark)

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11 Work out $(6.7 \times 10^4) \times (3.4 \times 10^{-8})$
Give your answer as an ordinary number.

(2 marks)

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12 Work out $\dfrac{0.03 \times 0.02}{0.008}$

Give your answer in standard form.

(3 marks)

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13 Work out $\dfrac{3.744 \times 10^9}{2.4 \times 10^5}$

Give your answer in standard form.

(2 marks)

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14 Work out the value of $(5 \times 10^3) \times (6 \times 10^7)$
Give your answer in standard form.

(2 marks)

15 (a) Write 0.000 054 376 in standard form. (1)

(b) Write 4.15×10^6 as an ordinary number. (1)

(c) Work out $\dfrac{4.1 \times 10^5 \times 7.3 \times 10^4}{2 \times 10^{-6}}$ (2)

(4 marks)

16 Write these numbers in order of size.
Start with the smallest number.

6.1×10^2 0.061×10^2 6100×10^{-4} 61

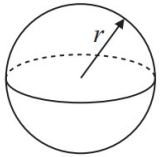
(2 marks)

17 A sphere has a radius of 6.4×10^6 metres.
Calculate the volume of this sphere.

Give your answer in standard form to 1 decimal place.

Volume of sphere $= \frac{4}{3}\pi r^3$

Surface area of sphere $= 4\pi r^2$



(3 marks)

18 A large rock has a weight of 1.2×10^4 grams.

Find, in standard form, the weight of 12 of these large rocks.

(2 marks)

19 Write these numbers in order of size.
Start with the smallest number.

3.5×10^2 0.035×10^5 350×10^{-2} 35×10^0

(2 marks)

20 The diameter of Neptune is 5.0×10^4 km
The diameter of Mars is 6.8×10^3 km
Calculate the difference between the diameter of Neptune and the diameter of Mars.
Give your answer in standard form.

(2 marks)

21 One electron has a mass of 9.1×10^{-31} grams.

Find the mass of 250 of electrons.

(2 marks)

22 The area of Australia is 7.7×10^6 km^2
The area of Cyprus is 9.3×10^3 km^2
How many times larger is Australia than Cyprus.
Give your answer to the nearest whole number.

(2 marks)

23 The area of the Pacific Ocean is 3.61×10^8 km^2
The area of the Atlantic Ocean is 8.51×10^7 km^2
Find the total area of the Pacific Ocean and the Atlantic Ocean.
Give your answer in standard form.

(2 marks)

24 The distance between Earth and Mars is 78 million kilometres.
The speed of light is 3×10^5 km/s

Calculate the time, in seconds, it takes for light to travel from Earth to Mars.
Give your answer in standard form.

(2 marks)